

ANNAPOLIS US NAVAL ACADEMY, MD, USA

WMO: 722158

Lat: 38.991N

Lon: 76.489W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-6.0	-17.8	0.8	-6.7	-15.5	1.0	-4.6	10.5	-0.4	9.2	0.3	3.9	290	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	32.9	24.6	31.7	24.3	30.4	23.8	26.4	30.5	25.7	29.6	25.0	28.9	3.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.4	28.6	24.4	19.4	28.0	23.8	18.7	27.5	82.0	30.3	78.8	29.4	76.0	28.9	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.6	6.9	DB	-10.9	35.8	2.7	1.8	-12.8	37.1	-14.4	38.1	-15.9	39.1	-17.9	40.4	(4)
(5)			WB	-12.2	27.9	2.5	1.0	-14.0	28.6	-15.4	29.2	-16.9	29.7	-18.7	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.5	3.4	7.4	13.6	18.8	23.8	26.3	25.3	22.3	15.8	9.6	5.2	(6)	
	DBStd	9.31	4.95	4.95	4.80	4.20	3.88	2.89	2.39	2.32	3.26	4.18	4.05	4.34	(7)	
	HDD10.0	769	237	192	109	13	0	0	0	0	4	55	159		(8)	
	HDD18.3	2220	491	418	339	152	43	2	0	0	4	101	263	407	(9)	
	CDD10.0	2437	4	7	29	121	273	414	507	475	369	184	43	10	(10)	
	CDD18.3	846	0	0	0	9	57	166	248	217	124	23	1	0	(11)	
	CDH23.3	6656	0	2	5	66	288	1212	2417	1817	765	83	2	0	(12)	
	CDH26.7	1952	0	0	0	17	62	331	842	518	169	12	0	0	(13)	
(14) Wind	WSAvg	3.0	3.1	3.3	3.5	3.5	3.0	2.8	2.6	2.5	2.7	2.9	2.9	2.8	(14)	
(15) Precipitation	PrecAvg	1168	89	67	108	92	94	107	113	114	109	98	80	94	(15)	
	PrecMax	1673	176	195	235	167	209	322	215	207	312	209	166	208	(16)	
	PrecMin	896	35	7	0	39	19	56	40	30	14	3	14	35	(17)	
	PrecStd	176	34	42	48	31	51	60	50	47	69	62	47	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	20.3	23.7	29.0	31.3	33.8	35.7	33.9	32.5	28.2	21.8	19.0	(19)	
		MCWB	14.2	14.3	15.5	18.0	23.5	24.2	25.9	24.8	24.8	22.6	17.3	15.6	(20)	
	2%	DB	13.5	15.7	19.4	24.9	28.2	31.7	33.4	32.1	30.1	25.7	18.9	15.5	(21)	
		MCWB	10.1	10.9	13.4	17.0	21.3	23.7	25.2	24.5	23.9	21.4	14.9	13.2	(22)	
	5%	DB	11.3	12.4	16.7	22.3	26.4	30.0	32.1	30.9	28.4	23.8	17.2	13.3	(23)	
		MCWB	8.4	8.8	11.9	16.1	20.6	23.0	24.7	24.1	23.1	20.2	13.6	11.0	(24)	
	10%	DB	9.1	9.9	14.0	20.1	24.6	28.5	30.8	29.2	27.2	22.3	15.7	11.2	(25)	
		MCWB	6.4	6.7	9.7	15.0	19.8	22.5	24.3	23.5	22.1	18.6	12.5	8.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.8	16.7	20.7	24.4	26.1	27.9	27.3	26.1	23.9	18.2	16.7	(27)	
		MCDB	18.8	19.1	21.5	23.5	28.8	30.7	32.5	30.6	30.4	26.3	20.3	18.4	(28)	
	2%	WB	11.4	12.0	14.6	18.4	22.9	25.2	26.8	26.0	24.9	22.6	16.5	13.8	(29)	
		MCDB	13.1	14.6	18.4	23.3	27.2	29.3	30.8	29.6	28.6	25.1	18.2	15.2	(30)	
	5%	WB	8.8	9.1	12.7	17.0	21.7	24.5	25.9	25.3	24.2	21.1	14.7	11.5	(31)	
		MCDB	10.7	11.6	16.0	21.3	25.3	28.4	30.0	29.0	27.6	23.3	16.6	12.7	(32)	
	10%	WB	6.7	6.9	10.6	15.6	20.7	23.8	25.3	24.6	23.3	19.3	13.1	9.5	(33)	
		MCDB	8.8	9.7	13.1	19.4	23.8	27.5	29.3	28.2	26.3	21.5	14.9	10.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.4	7.8	8.6	7.7	7.3	7.0	6.7	6.6	7.2	7.2	6.5	(35)	
		MCDBR	10.4	11.4	11.8	12.1	9.3	8.8	8.4	8.0	7.5	8.0	9.2	9.3	(36)	
	5% WB	MCWBR	8.2	8.3	7.7	6.3	4.9	4.4	3.9	3.8	3.8	4.9	6.6	8.1	(37)	
		MCWBR	9.8	10.7	10.9	10.8	8.3	7.7	7.3	7.0	6.8	6.8	8.0	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.311	0.321	0.360	0.399	0.443	0.496	0.518	0.512	0.424	0.362	0.326	0.311	(40)	
		taud	2.481	2.441	2.389	2.300	2.210	2.107	2.059	2.062	2.282	2.457	2.516	2.522	(41)	
	Ebn at Noon		865	904	897	878	842	796	773	766	819	845	841	839	(42)	
		Edh at Noon	79	94	110	127	142	158	164	160	121	92	76	71	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.08	2.93	3.90	4.97	5.60	6.20	6.10	5.42	4.40	3.28	2.37	1.83	(44)	
	RadStd		0.16	0.33	0.31	0.39	0.67	0.47	0.29	0.31	0.49	0.36	0.20	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (80 neighbors)		+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+79	(47)	

Nomenclature: See separate page